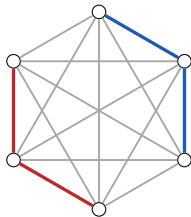


Ramsey Numbers

Section 1.8

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



Today's lecture

1. Classical Ramsey numbers

Party problem; definition of $R(p, q)$.

2. First exact values

$R(3, 3) = 6$ and $R(3, 4) = 9$.

3. General bounds

Recurrence; binomial upper bound.

4. Graph Ramsey numbers

From $R(K_p, K_q)$ to $R(G, H)$.

The party problem

You are going to a party. Some pairs of people already know each other, and some pairs of people are strangers.

Assume that for every pair of people, exactly one of these two possibilities holds:

they know each other

or

they are strangers

Question

How many people are needed to guarantee either
three mutual acquaintances or **three mutual strangers**?

The party problem as an edge-colouring problem

Build the complete graph K_n

- each person is a vertex;
- each pair of people is joined by an edge.

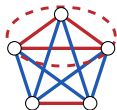
Colour every edge

- **red**: the two people know each other;
- **blue**: the two people are strangers.

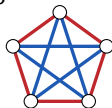
Equivalent question

How large must n be so that every red–blue edge-colouring of K_n contains a **monochromatic** copy of K_3 ?

Two edge-colourings of K_5



Contains a monochromatic K_3



Contains no monochromatic K_3

Now, some definitions

2-colouring

A **2-colouring** of the edges of a graph assigns one of two colours to every edge.

Classical Ramsey number

Let p and q be positive integers. The **Ramsey number** $R(p, q)$ is the smallest integer n such that every 2-colouring of the edges of K_n contains either

a red K_p or a blue K_q

as a subgraph.

In particular, $R(3, 3)$ answers the party problem.

Basic Question

Question

Suppose $n > R(p, q)$. Then every 2-colouring of the edges of K_n must contain either a red K_p or a blue K_q . Why?

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Suppose $n > R(p, q)$. Then every 2-colouring of the edges of K_n must contain either a red K_p or a blue K_q . Why?

Answer

Removing m vertices from K_n leaves the complete graph K_{n-m} . Take

$$m = n - R(p, q).$$

The remaining vertices span a copy of $K_{R(p,q)}$. Any 2-colouring of K_n restricts to a 2-colouring of this subgraph, which contains a red K_p or a blue K_q by the definition of $R(p, q)$.

Next Question

Question

What is the value of $R(1, 3)$?

Next Question

Question

What is the value of $R(1, 3)$?

Answer

$$R(1, 3) = 1.$$

The graph K_1 has one vertex and no edges. Thus, in every 2-colouring of K_1 , the single vertex already forms a red K_1 : there are no edges that could fail to be red.

Therefore $n = 1$ satisfies the definition, and it is the smallest possible positive integer.

Follow-Up Question

What about the value of $R(1, k)$?

Next Question

Question

What is the value of $R(2, 4)$?

Next Question

Question

What is the value of $R(2, 4)$?

Claim

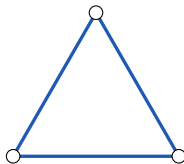
$$R(2, 4) = 4.$$

We will prove this claim in two steps.

Step 1. Show that $R(2, 4) > 3$ by exhibiting a 2-colouring of K_3 with no red K_2 and, trivially, no blue K_4 .

Step 2. Show that $R(2, 4) \leq 4$ by proving that every 2-colouring of K_4 contains a red K_2 or a blue K_4 .

Showing $R(2, 4) > 3$



Why this colouring works

Every edge is blue, so there is no red K_2 .

The graph has only three vertices, so it cannot contain a blue K_4 .

Conclusion

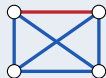
Thus, $R(2, 4) > 3$.

Showing $R(2, 4) \leq 4$

Proof

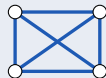
Consider any 2-colouring of the edges of K_4 . There are two cases.

Case 1: a red edge occurs



That edge is a red K_2 .

Case 2: no red edge occurs



The whole graph is a blue K_4 .

Conclusion

Every 2-colouring of K_4 contains a red K_2 or a blue K_4 . Therefore, $R(2, 4) \leq 4$.

Follow-Up Question

Question

What about the value of $R(2, n)$?

The value of $R(2, n)$

Question

What about the value of $R(2, n)$?

Claim

$$R(2, n) = n.$$

The same argument gives both bounds

For $R(2, n) > n - 1$, colour every edge of K_{n-1} blue. There is no red K_2 , and there are too few vertices for a blue K_n .

For $R(2, n) \leq n$, consider any 2-colouring of K_n . A red edge gives a red K_2 ; if there is no red edge, then all edges are blue, giving a blue K_n .

Computing Another Ramsey Number

Claim

$$R(3, 3) = 6.$$

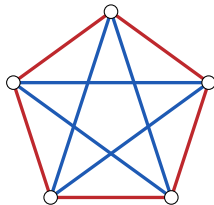
Proof

Again, we will prove the claim in two steps.

Step 1. Show that $R(3, 3) > 5$ by exhibiting a 2-colouring of K_5 with neither a red K_3 nor a blue K_3 .

Step 2. Show that $R(3, 3) \leq 6$ by proving that every 2-colouring of K_6 contains a red K_3 or a blue K_3 .

Showing $R(3, 3) > 5$

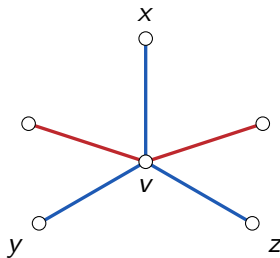


Why this colouring works

Notice that this 2-colouring of K_5 contains no red or blue K_3 subgraph.

Therefore, $R(3, 3) > 5$.

Showing $R(3, 3) \leq 6$



The pigeonhole idea

Choose any vertex v of K_6 . Its five incident edges are coloured red or blue, so at least three have the same colour.

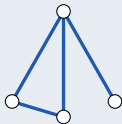
Without loss of generality, suppose v is joined to x, y, z by **blue** edges.

Showing $R(3, 3) \leq 6$

What happens among x, y, z ?

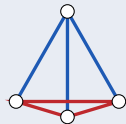
Recall that vx , vy , and vz are blue. There are two cases.

Case 1: a blue edge occurs



The edge xy closes a blue K_3 .

Case 2: no blue edge occurs



Then xyz is a red K_3 .

Conclusion

Every 2-colouring of K_6 contains a red or blue K_3 . Thus, $R(3, 3) \leq 6$.